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SUMMARY

Pre-final year undergraduate at IITRAM with **3 published IEEE papers** and **3 manuscripts under review** in automated sleep disorder detection, biomedical signal processing, and medical image analysis. Research focuses on building clinically interpretable deep learning systems that neurologists can audit and deploy in real environments — combining EEG/PPG analysis, explainable AI, and Vision Transformer architectures. Alongside research, experienced building and shipping full-stack AI products in production: facial recognition systems, LLM-powered SaaS platforms, telemedicine infrastructure, and computer vision pipelines on edge hardware. Awarded the **Undergraduate Research Award twice** by IITRAM; **1st Runner-Up at IIT Bombay** national hackathon; winner or runner-up across **4 additional hackathons**.

EDUCATION

Institute of Infrastructure Technology Research and Management (IITRAM) *Jul 2023 – Aug 2027*
Bachelor of Technology – Computer Engineering *CGPA: 7.53/10*

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Linear Algebra, Probability & Statistics, Digital Signal Processing, Machine Learning, Computer Vision

PUBLICATIONS

Published – IEEE

- **Automated Identification of Cyclic Alternating Patterns of Sleep Using Fusion of VGG16 and Vision Transformer**
IEEE Access, 2025 DOI: 10.1109/ACCESS.2025.3571145
Designed a hybrid CNN-ViT architecture fusing spatial feature extraction from VGG16 with global sequence modelling from Vision Transformers to automate detection of Cyclic Alternating Patterns (CAP) in overnight polysomnographic EEG recordings. Demonstrated improved detection accuracy over baseline CNN-only approaches on publicly available sleep EEG datasets.
- **PPG-Based Accurate Insomnia Detection System Using CNN With Self-Attention Mechanism and GRU**
IEEE Access, 2025 DOI: 10.1109/ACCESS.2025.3598863
Proposed a novel deep learning architecture combining convolutional layers for local PPG feature extraction, a self-attention mechanism for long-range dependency modelling, and Gated Recurrent Units for temporal sequence learning. System achieves high-accuracy insomnia detection from wrist-worn PPG signals, enabling non-invasive, wearable-compatible sleep disorder screening.
- **Automated Sleep Stage Classification Using Machine Learning and Signal Processing Techniques**
IEEE International Conference on AI and Machine Vision (AIMV), 2025 DOI: 10.1109/AIMV66517.2025.11203559
Developed a biorthogonal wavelet decomposition pipeline for feature extraction from polysomnographic EEG signals, combined with ensemble machine learning classifiers for multi-class sleep stage classification. Evaluated on standard benchmark datasets against established sleep scoring criteria.

Manuscripts Under Review

- **Explainable Multimodal Deep Learning Framework for K-Complex Detection using EEG Signals**
Under Review
Proposes a multimodal deep learning framework for automated K-complex detection in EEG signals, integrating Grad-CAM and SHAP-based explainability to generate clinician-readable saliency maps. Addresses the interpretability gap in clinical sleep scoring AI, enabling neurologists to audit and validate model decisions before deployment.
- **Automatic Sleep Spindle Detection Using Separable Convolutional Neural Network With EEG Signals**
Under Review
Introduces a computationally efficient separable CNN architecture for automated sleep spindle detection, significantly reducing parameter count and inference time compared to standard CNN baselines while maintaining classification accuracy on overnight EEG recordings.
- **An Attention-Based Denoising Model for Diffusion Weighted Imaging**
Under Review
Develops an attention-gated denoising network for improving signal quality in diffusion-weighted MRI scans, targeting clinical scenarios with low SNR acquisitions. Demonstrates noise reduction performance while preserving diagnostically relevant diffusion tensor features.

EXPERIENCE

Machine Learning Engineer Intern

Insolare Pvt. Ltd., Ahmedabad – Remote

Apr 2025 – Present

Python, Computer Vision, Deep Learning, GPS/GIS

- Architected and deployed a passive facial recognition attendance system with active anti-spoofing (liveness detection), achieving **98%+ verification accuracy** across multiple operational workshop sites with zero manual intervention.
- Designed and trained a drone-based solar panel defect detection pipeline using object detection and image segmentation, **reducing manual inspection time by 75%** and enabling automated classification of physical and electrical panel damage.
- Built a geospatial monitoring dashboard integrating live GPS telemetry data to visualise solar field health, track equipment locations, and optimise maintenance crew routing.
- Developed a Solar Panel Operations and Management platform enabling real-time monitoring of panel performance metrics, input/output power tracking, and condition-based maintenance scheduling.
- Deployed all systems to production environments actively used in Insolare operations, providing end-to-end exposure from model training through containerised deployment and live monitoring.

Software Development Engineer Intern

Garnet AI, Dublin, Ireland – Remote

May 2025 – Jul 2025

React, Node.js, PostgreSQL, Docker, AWS, OpenAI API

- Led full-stack development of a SaaS vendor onboarding platform for enterprise B2B clients, **reducing onboarding time by 40%** through automated document workflows and self-serve vendor portals.
- Integrated GPT-powered automation pipelines for compliance documentation generation, regulatory requirement extraction, and vendor verification, **cutting manual processing workload by 60%**.
- Built and maintained secure REST APIs using Node.js and PostgreSQL with JWT-based authentication, role-based access control, and audit logging for enterprise clients.
- Established CI/CD pipelines using Docker, GitHub Actions, and AWS (EC2 + S3), enabling zero-downtime deployments and automated testing across staging and production environments.
- Collaborated with a cross-functional remote team across product, design, and backend engineering to ship features on a 2-week sprint cycle.

Research Intern

Indian Institute of Technology Delhi – Remote

Jan 2025 – Jun 2025

PyTorch, EEG Signal Processing, Explainable AI, MRI

- Designed a separable convolutional neural network for automated sleep spindle detection in overnight EEG recordings, achieving **98.08% detection accuracy** on benchmark polysomnographic datasets with significantly reduced computational cost versus standard CNN baselines.
- Developed an explainable multimodal deep learning framework for K-complex detection achieving **99.58% classification accuracy**, incorporating Grad-CAM and SHAP visualisations to produce clinician-readable saliency maps for model audit and clinical validation.
- Applied gradient-based and perturbation-based explainability methods (Grad-CAM, SHAP) to produce interpretable visualisations of EEG feature importance, directly supporting neurologist review of automated sleep scoring outputs.
- Designed and implemented an attention-gated denoising network for diffusion-weighted MRI, improving SNR in low-quality clinical scan acquisitions while preserving diagnostically relevant tensor features.
- Managed complete ML pipelines from raw EEG/PPG signal preprocessing (bandpass filtering, epoch segmentation, artefact rejection) through model training, hyperparameter tuning, and cross-validation on clinical datasets.
- Produced 2 peer-reviewed manuscripts currently under review at IEEE journals, extending 2 published IEEE Access papers from the same research programme.

Machine Learning Intern

MarkX AI Labs, Ahmedabad – Remote

Nov 2024 – Dec 2024

Python, Deep Learning, Algorithmic Trading, Telegram API

- Developed LSTM and gradient-boosted ML models for short-term market trend prediction using historical OHLCV data and technical indicators.
- Built a Telegram-based trading assistant delivering automated real-time buy/sell signals, stop-loss recommendations, and portfolio analytics to subscribed users.
- Designed automated signal generation pipelines processing live market data through trained models to produce actionable investment recommendations with configurable risk thresholds.

AI Developer Intern

Blix Education, Ahmedabad – Remote

Sep 2024 – Nov 2024

Python, Computer Vision, OpenCV, Raspberry Pi, Edge ML

- Developed an autonomous navigation prototype on Raspberry Pi integrating a camera-based perception stack with lane detection, obstacle recognition, and PID-based steering control for real-time vehicle movement.
- Implemented computer vision modules using OpenCV for lane boundary segmentation and ultrasonic sensor fusion for obstacle avoidance, optimised for low-latency inference on edge hardware.

- Conducted model compression and quantisation to fit inference pipeline within Raspberry Pi memory and CPU constraints while maintaining real-time navigation performance.
- Designed modular software architecture separating perception, decision, and actuation layers to allow independent testing and iterative improvement of each navigation component.

Data Science Intern

Jun 2024 – Jul 2024

Yads Technology Pvt. Ltd. – Remote

Python, BeautifulSoup, Pandas, Data Analysis

- Built automated web scraping pipelines using Python, BeautifulSoup, and Scrapy to systematically collect structured datasets from multiple online sources at scale.
- Designed data cleaning, deduplication, and validation workflows using Pandas to standardise raw scraped data for downstream analytics consumption.
- Performed exploratory data analysis and produced stakeholder-facing reports and visualisations (Matplotlib, Seaborn) to communicate trends and operational insights.
- Developed reusable data extraction utilities that significantly reduced manual data collection effort and improved consistency of internal analytics datasets.

PROJECTS

CareerBuddy – AI-Powered Career Guidance Platform

GitHub

Stack: Next.js, NestJS, MongoDB, TypeScript, Docker, Socket.IO, JWT, AI/NLP

- Engineered a production-grade full-stack SaaS platform supporting three user roles (Students, Counselors, Admins) with role-based dashboards, secure JWT authentication, and granular permission controls.
- Built an AI-driven resume intelligence module that parses uploaded PDFs, extracts skills using NLP, performs ATS keyword gap analysis, and generates personalised feedback reports for students.
- Implemented an intelligent job-matching pipeline that ranks opportunities against candidate skill profiles, experience history, and stated preferences using cosine similarity and weighted scoring.
- Developed real-time communication infrastructure using Socket.IO for instant counselor-student messaging, live application status notifications, and session scheduling.
- Designed a scalable NestJS backend with MongoDB supporting resume versioning, application tracking, analytics dashboards, and admin reporting; containerised with Docker for cloud deployment.
- Architected modular service boundaries allowing independent scaling of resume parsing, job recommendation, and messaging services across cloud environments.

AI-Powered TeleMedicine Platform

GitHub

Stack: React, NestJS, Flask, MongoDB, Python, ML, WebRTC, Socket.IO, OpenCV, Tesseract

- Built end-to-end during Smart India Hackathon: a unified digital healthcare ecosystem integrating telemedicine, AI health assistance, e-pharmacy, and hospital discovery into a single platform.
- Implemented secure multi-role authentication (patients and healthcare professionals) with JWT-based access control, session management, and profile systems.
- Built real-time video consultation infrastructure using WebRTC and Socket.IO, enabling low-latency virtual appointments with instant in-session messaging between patients and doctors.
- Integrated ML-based health assistant modules: TF-IDF symptom matching for preliminary diagnosis, OCR-based prescription extraction using Tesseract and OpenCV, and voice-enabled patient interactions.
- Developed complete appointment scheduling, digital prescription management, and medicine ordering systems with inventory tracking and pharmacy integration.
- Designed Flask-based AI microservices architecture separating ML inference from the main Node.js application, enabling independent model updates without service disruption.

UrbanPulse – Smart City Command Center

GitHub

Stack: React, Vite, Leaflet, OpenStreetMap, Recharts, Data Analytics

- Designed and developed a multi-module smart city monitoring platform covering traffic analytics, environmental indicators (AQI, noise, temperature), utility management, and emergency response dashboards.
- Implemented live simulation engines for dynamic city-health monitoring with configurable alert thresholds, incident logging, and cross-region operational metrics tracking.
- Integrated Leaflet and OpenStreetMap for geospatial visualisation: interactive city maps with sensor overlays, real-time incident markers, and location-aware analytics across multiple metropolitan regions.
- Built interactive Recharts dashboards enabling trend analysis, anomaly detection, and exportable reporting for urban planners and emergency services operations.

- Designed a scalable frontend architecture with modular component boundaries supporting future integration of live IoT sensor APIs and streaming data pipelines.

KhetiBuddy – Smart Agriculture Assistance System

GitHub

Stack: Python, Scikit-Learn, Pandas, Matplotlib, Data Analytics

- Developed an intelligent crop recommendation system using Random Forest and gradient-boosted classifiers trained on soil composition, climate, and irrigation parameters to suggest optimal crops for given conditions.
- Built yield prediction models processing multi-variate agricultural input features (soil pH, rainfall, humidity, fertiliser) to generate season-specific productivity estimates for farmers.
- Designed analytical workflows that ingest raw environmental sensor and historical yield datasets, apply feature engineering, and output actionable farming recommendations via a Python-based interface.
- Created data visualisation dashboards displaying crop performance trends, resource utilisation analytics, and seasonal pattern analysis to support data-driven farm management decisions.
- Developed a modular codebase allowing plug-in integration of additional prediction models (disease forecasting, weather intelligence) without architectural rework.

POSITIONS OF RESPONSIBILITY

Coordinator, Career Development Centre (CDC), IITRAM

Jul 2025 – Present

- Managing and coordinating a 34-member multidisciplinary team across outreach, operations, content creation, and industry engagement verticals for IITRAM's central placement and career body.
- Organised and conducted 6+ workshops, mock interview sessions, resume review drives, and career guidance programs reaching 300+ students across all undergraduate batches.
- Established and maintained partnerships with industry recruiters and HR professionals to facilitate internship drives, placement talks, and campus hiring processes.
- Coordinating placement logistics between students, faculty placement coordinators, and visiting company representatives to ensure smooth execution of hiring cycles.
- Developed content and communication strategy for CDC's student-facing channels, improving student engagement and awareness of career development resources.

Team Lead, Alumni Relations, IITRAM

Jul 2025 – Present

- Leading a team of 50+ student volunteers responsible for planning, coordinating, and executing alumni engagement initiatives for the institute.
- Organised alumni interaction sessions, panel discussions, and mentorship programmes connecting current students with alumni across industry, academia, and entrepreneurship.
- Coordinating with distinguished alumni to build long-term institutional relationships that translate into mentorship pipelines, referral networks, and guest lecture series.
- Managing full event lifecycle for alumni outreach: venue coordination, communication workflows, stakeholder scheduling, and post-event follow-up.

AI/ML Team Lead, Tekstra Coding Club, IITRAM

Jan 2025 – Aug 2025

- Led the AI/ML vertical of IITRAM's coding community, designing and delivering structured learning tracks covering machine learning fundamentals, deep learning, and applied AI projects.
- Organised technical workshops, hackathons, and hands-on lab sessions attended by 100+ students; mentored junior members through project-based learning initiatives.
- Curated and published AI/ML newsletters and resource guides to maintain active knowledge sharing within the technical community.
- Facilitated peer mentorship connecting seniors with first and second-year students to build foundational programming and ML skills.

Senior Member / Core Member, Tekstra Coding Club, IITRAM

Oct 2023 – Jan 2025

- Contributed to competitive programming, web development, and AI sessions as a core technical member before transitioning to the AI/ML leadership role.

Campus Mantri, GeeksforGeeks

Dec 2024 – Jun 2025

- Served as institutional representative for GeeksforGeeks, promoting competitive programming resources, DSA learning tracks, and technical article engagement among students on campus.
- Organised coding awareness sessions and drove participation in GFG challenges, increasing campus community engagement with the platform.

Campus Ambassador, Unstop

Oct 2024 – May 2025

- Promoted hackathons, case competitions, and skill-building challenges on campus; helped connect students with national-level competition opportunities and professional development tracks.

ACHIEVEMENTS & HONORS

- Awarded the **Undergraduate Research Award (URA)** on **two separate occasions** by IITRAM in recognition of outstanding research contributions in Deep Learning and Biomedical Signal Processing – one of the few undergraduates to receive this recognition twice.
- **1st Runner-Up** at a national-level hackathon hosted by **IIT Bombay**, competing against multidisciplinary student teams from premier institutions across India.
- Winner or runner-up in **3 additional hackathons** spanning domains of Artificial Intelligence, Healthcare Technology, and Full-Stack Software Development.
- **3 IEEE publications** as an undergraduate in a specialised and competitive domain (biomedical signal processing / sleep analytics), with **3 further manuscripts currently under review** – reflecting a sustained independent research pipeline unusual at the undergraduate level.
- Selected for a **Research Internship at IIT Delhi** (remote) in the area of EEG signal analysis and explainable AI for clinical neuroscience applications.
- Maintaining an active open-source portfolio on **GitHub** spanning production ML systems, full-stack AI platforms, healthcare AI, smart-city dashboards, and agricultural ML tools.

TECHNICAL SKILLS

Expert (publication-backed): PyTorch, EEG/PPG Signal Processing, Vision Transformers, CNNs, Separable Convolutions, Explainable AI (Grad-CAM, SHAP), Biomedical Deep Learning, Medical Image Analysis, Sleep Stage Classification

Proficient (production use): TensorFlow, Scikit-Learn, Hugging Face Transformers, FastAPI, Flask, React, Next.js, NestJS, Node.js, Docker, GitHub Actions, AWS (EC2, S3), PostgreSQL, MongoDB, WebRTC, Socket.IO

Familiar: LangChain, LLMs, RAG, Reinforcement Learning, MLflow, Prefect, FAISS, ChromaDB, MySQL, Supabase, Vercel, Railway

Languages: Python, JavaScript, TypeScript, SQL, C++, MATLAB, HTML/CSS

Research Areas: Biomedical Signal Processing, EEG Analysis, PPG Analysis, Sleep Stage Classification, K-Complex Detection, Sleep Spindle Detection, Diffusion-Weighted Imaging, Explainable AI, Clinical ML, Healthcare AI

CERTIFICATIONS

- **Machine Learning Specialization** (3 courses) – DeepLearning.AI / Stanford Online
- **Deep Learning Specialization** (5 courses: Neural Networks, Improving DNNs, Structuring ML Projects, CNNs, Sequence Models) – DeepLearning.AI
- **IBM Data Science Professional Certificate** (10-course program) – IBM
- **Google AI Essentials** – Google
- **Building Generative AI-Powered Applications with Python** – IBM
- **Google IT Automation with Python** – Google